

TECHNICAL DOCUMENT

SpaceOBC1

Space On-Board Computer System

Technical Operations Manual

Architectural and Operational Specification

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Table of Contents

1. System Introduction

1.1 Document Purpose

This document constitutes the complete technical operational specification for the SpaceOBC1 (Space On-Board Computer) system. Its primary purpose is to provide a detailed description of the software architecture, configuration parameters, temporal dependencies, and operational dynamics of the system, in order to support development, maintenance, testing, and mission operations activities.

This document has been prepared with particular attention to knowledge transfer, thus representing a fundamental resource for onboarding new technical personnel and ensuring operational continuity of the system.

1.2 Scope of Application

The SpaceOBC1 system represents a complex software architecture specifically designed for the management and control of a space on-board computer (On-Board Computer). The system was conceived to operate in a critical environment where reliability, timing precision, and efficient use of computational resources are fundamental requirements for mission success.

The system architecture is based on a distributed execution model that coordinates multiple specialized services, each responsible for specific operational functions. These services orchestrate the execution of interdependent tasks that must comply with strict timing constraints and well-defined precedence relationships.

1.3 Intended Audience

This document is intended for: software engineers responsible for system development and maintenance, mission operators responsible for operational monitoring and control, quality assurance teams for verification and validation activities, training personnel for knowledge transfer programs, and technical management for project activity supervision.

2. Global Configuration and Operational Parameters

2.1 Operational Time Window

The SpaceOBC1 system operates within a well-defined time window extending from the initial instant, identified as time zero, up to time unit thirty-three. This operational window of thirty-three time units represents the period during which all scheduled operations must be started, executed, and completed.

The choice of this specific duration is closely linked to the requirements of the space mission and the orbital characteristics of the vehicle, ensuring that all critical operations can be completed within the constraints imposed by orbital dynamics and communication windows with ground stations.

2.2 System Parameters

Parameter	Value	Description
System Name	SpaceOBC1	Unique instance identifier
Operational Window (t_max)	33 t.u.	Maximum operational cycle duration
Max Resources (r_max)	8	Simultaneous processing units
Active Services	7	Number of service modules
Total Tasks	11	Number of operational tasks

The availability of eight resources allows a significant degree of parallelism in task execution, enabling the system to effectively exploit the processing capabilities of the on-board computer. However, this constraint also requires careful planning to avoid resource contention situations that could compromise the timely completion of critical operations.

3. Modular Service Architecture

3.1 Architecture Overview

The SpaceOBC1 system architecture is structured into seven main services, each designed to manage a specific functional domain of on-board operations. This modular organization enables a clear separation of responsibilities and facilitates both maintenance and system evolution over time.

ID	Service	Primary Responsibility
1	CoreService	Fundamental operations management and system initialization
2	ControlSyncService	Real-time control and component synchronization
3	AnalysisService	Data analysis and operational integrity verification
4	ProcessingService	Data processing pipeline (pre/post processing)
5	OrbitService	Orbital control and vehicle orbit management
6	MonitorService	Continuous system status monitoring
7	SafeService	Safety management and safe mode procedures

3.2 Detailed Service Descriptions

3.2.1 CoreService (ID: 1)

The CoreService represents the foundation of the entire operational architecture. This service has the critical responsibility of managing the system's fundamental operations through two essential tasks: core initialization (INIT_CORE) and communication protocol configuration (COMM_PROTO).

The importance of this service cannot be underestimated, since virtually all other system operations depend directly or indirectly on the correct completion of its functions. CoreService establishes the necessary operational baseline, configuring essential system components and preparing the communication infrastructure.

3.2.2 ControlSyncService (ID: 2)

The ControlSyncService handles real-time control management and synchronization among system components. Through its two main tasks, CTRL_EXEC and SYNC_PROC, this service ensures that control operations are executed with the required frequency and precision, and that all subsystems maintain accurate time synchronization.

The repetitive nature of these tasks reflects the need for continuous control and periodic synchronization throughout the entire operational window.

3.2.3 AnalysisService (ID: 3)

The AnalysisService is dedicated to in-depth system data analysis and operational integrity verification. This service coordinates two complementary tasks: ANALYSIS_FULL, which performs a complete and detailed analysis of system state, and VERIFY_QUICK, which carries out rapid validation checks.

Combining deep analysis with quick verification allows the system to maintain an optimal balance between analytical completeness and timely detection of potential anomalies.

3.2.4 ProcessingService (ID: 4)

The ProcessingService manages the entire data processing pipeline through two distinct but closely related phases. The pre-processing phase, handled by the PREPROCESS_SVC task, prepares data for subsequent processing by applying transformations, filtering, and normalization. The post-processing phase, handled by POSTPROCESS_SVC, completes the process by applying further transformations, aggregations, and preparing final results.

3.2.5 OrbitService (ID: 5)

The OrbitService has particular relevance in the context of space operations, as it is responsible for all functions related to orbital control and vehicle orbit management. This service coordinates two specialized tasks: ORBIT_CTRL, which performs active orbit control, and ORBIT_MGMT, which manages broader orbital operations.

3.2.6 MonitorService (ID: 6)

The MonitorService provides continuous system status monitoring capabilities through the MONITOR_SYS task. The distinctive feature of this service is its ability to run up to three simultaneous instances of the monitoring task, providing redundancy that significantly increases system reliability in detecting and reporting operational anomalies.

3.2.7 SafeService (ID: 7)

The SafeService manages all aspects related to system safety and safe mode procedures through the SAFE_DIR task. This service represents the last line of defense in critical situations, implementing the necessary safety directives to protect the spacecraft and ensure mission continuity even in the presence of significant anomalies.

4. Detailed Description of Operational Tasks

4.1 Complete Task Registry

ID	Task	Code	Service	Duration	Resources	Reps	Interval
1	INIT_CORE	initcor	1	14	1	1	-
2	COMM_PROTO	comprt	1	16	1	1	-
3	CTRL_EXEC	ctlexe	2	3	1	3	8
4	SYNC_PROC	syncpr	2	6	1	3	8
5	ANALYSIS_FULL	anlyfl	3	14	2	1	-
6	VERIFY_QUICK	vrfyqk	3	7	1	1	-
7	PREPROCESS_SVC	prepsv	4	9	2	1	-
8	POSTPROCESS_SVC	pstpsv	4	12	2	1	-
9	ORBIT_CTRL	orbctl	5	15	2	1	-
10	ORBIT_MGMT	orbmgt	5	8	1	1	-
11	MONITOR_SYS	monsys	6	12	1	3	-
12	SAFE_DIR	safdir	7	4	1	1	-

Legend: Duration = time units; Resources = required computational units; Reps = number of repetitions; Interval = interval between repetitions (t.u.)

5. Temporal Constraints and Dependency System

5.1 Start-to-Start Precedence Model

The SpaceOBC1 system implements a sophisticated temporal constraint system based on the concept of start-to-start precedence. This type of constraint specifies that a destination task can begin execution only after a given delay has elapsed since the start of the source task execution.

It is important to note that these constraints do not require completion of the source task, but only that the specified time has passed since its start. This feature enables a greater degree of parallelism in execution, optimizing the use of available resources.

5.2 Dependency Matrix

Source Task	Destination Task	Delay (t.u.)	Type
INIT_CORE	CTRL_EXEC	2	SS
COMM_PROTO	CTRL_EXEC	2	SS
INIT_CORE	ANALYSIS_FULL	4	SS
COMM_PROTO	ANALYSIS_FULL	4	SS
CTRL_EXEC	SYNC_PROC	2	SS
ANALYSIS_FULL	VERIFY_QUICK	8	SS
VERIFY_QUICK	PREPROCESS_SVC	6	SS
VERIFY_QUICK	ORBIT_CTRL	6	SS
VERIFY_QUICK	ORBIT_MGMT	6	SS
PREPROCESS_SVC	POSTPROCESS_SVC	2	SS
INIT_CORE	MONITOR_SYS	10	SS

Legend: SS = Start-to-Start (destination task can start after the specified delay from source task start)

5.3 Critical Dependency Analysis

The dependency chain begins with the two fundamental initialization tasks: INIT_CORE and COMM_PROTO. Both tasks act as sources for multiple constraints, reflecting their critical role in system startup.

VERIFY_QUICK represents a particularly interesting branching point in the system architecture. Three different tasks depend on its start, all with the same delay of six time units: PREPROCESS_SVC, ORBIT_CTRL, and ORBIT_MGMT. This configuration enables the

simultaneous start of three parallel operational paths once quick verification has confirmed data validity and nominal system status.

6. Final Considerations and Recommendations

6.1 Architecture Summary

The SpaceOBC1 system represents a sophisticated example of software architecture for critical space applications. Its design reflects a deep understanding of operational requirements, timing constraints, and robustness needs typical of space missions.

The balance between parallelism and sequential dependencies, careful management of computational resources, and the presence of monitoring and safety mechanisms demonstrate a mature approach to critical system design.

6.2 Modular Architecture Benefits

The modularity of the architecture, with its clear separation into specialized services, facilitates not only system understanding but also future maintenance and evolution. The ability to modify or extend individual services without impacting the entire architecture represents a significant advantage in adapting the system to changing mission requirements or integrating new functionality.

6.3 Knowledge Transfer Recommendations

For effective knowledge transfer, it is recommended to follow a structured training path that includes: understanding global parameters and the operational window, studying the service architecture starting from CoreService, detailed analysis of individual tasks and their characteristics, mapping temporal dependencies and understanding the start-to-start model, simulation of operational flow with particular attention to parallelization points, and finally studying safety procedures and the role of SafeService.

Appendix A: Glossary

Term	Definition
OBC	On-Board Computer - Spacecraft on-board computer
t.u.	Time Unit - Base unit for time measurement in the system
r_max	Maximum number of computational resources allocable simultaneously
t_max	Maximum duration of operational window in time units
Start-to-Start (SS)	Type of precedence constraint based on task start rather than completion
Safe Mode	Safety operational mode activated in case of critical anomalies
Periodic Task	Task configured to execute multiple repetitions at regular intervals
Pipeline	Processing sequence where output of one phase becomes input of the next

Appendix B: Reference Documents

Code	Document Title	Version
REF-001	SpaceOBC1 System Requirements Specification	[To be filled]
REF-002	Interface Control Document - Ground Segment	[To be filled]
REF-003	Verification and Validation Plan	[To be filled]
REF-004	Mission Operations Manual	[To be filled]

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